

JESS Thermodynamic Database v8.9

Radium

24-Jan-23

Reaction No. 4186							
EDTA-4 + Ra+2 = Ra+2_EDTA-4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	KNO3	lgK:7.7(2SF)	0	MSC	903
2	20	0.1	NaCl	lgK:7.07(3SF)	3	EIU	903
3	20	0.1	NaCl	lgK:7.1(0.06SD)	3	MIX	903
4	20	0.1	Unknown	lgK:7.1(2SF)	2	ENS	773
5	23			lgK:7.12(3SF)	3	MIX	903
6	25	0	Inf. Dilution	lgK:8.879(4SF)	1	EOR	
7	25	0	Inf. Dilution	lgK:9.13(0.07SD)	5	EDH	29889
8	25	0	NaCl/m	lgK:9.11(0.09SD)	3	EDH	29889
Note (8): Strangely, different from Inf Dil value in abstract							
9	25	0.1	Cl-1 salt	lgK:7.4(2SF)	4	MIX	903
10	25	0.1	KNO3	lgK:7.5(2SF)	3	MSC	903
11	25	0.1	NaClO4	lgK:7.5(2SF)	3	MDS	906
12	25	0.22	NaCl/m	lgK:6.96(0.20SD)	3	MIX	29889
13	25	0.51	NaCl/m	lgK:6.60(0.08SD)	3	MIX	29889
14	25	1.02	NaCl/m	lgK:6.42(0.10SD)	3	MIX	29889
15	25	2.09	NaCl/m	lgK:6.60(0.10SD)	3	MIX	29889
16	25	2.64	NaCl/m	lgK:6.63(0.08SD)	3	MIX	29889

Reaction No. 4857							
Ra+2 + 2<e-1> = Ra(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-98.39(4SF)	4	CRV	32224
2	25	0	Inf. Dilution	dG:134.2(0.1SD)	3	ENS	3006
3	25	0	Inf. Dilution	dG:134.2(4SF)	4	CRV	5223
4	25	0	Inf. Dilution	dG:134.2(4SF)	4	CRV	6488
5	25	0	Inf. Dilution	dG:561.5(4SF) kJ	5	CRV	7421
6	25	0	Inf. Dilution	dG:134.200(6SF)	4	CRV	9029
7	25	0	Inf. Dilution	dG:134.20(5SF)	4	CRV	10174

8	25	0	Inf. Dilution	dG:534.(3SF) kJ	0	CRV	22524
9	25	0	Inf. Dilution	dG:561.5(4SF) kJ	4	EFE	32151
10	50	0	Inf. Dilution	dG:134.51(5SF)	2	CRV	10174
11	75	0	Inf. Dilution	dG:134.81(5SF)	2	CRV	10174
12	100	0	Inf. Dilution	dG:135.08(5SF)	2	CRV	10174
13	125	0	Inf. Dilution	dG:135.34(5SF)	2	CRV	10174
14	150	0	Inf. Dilution	dG:135.59(5SF)	2	CRV	10174
15	175	0	Inf. Dilution	dG:135.81(5SF)	2	CRV	10174
16	200	0	Inf. Dilution	dG:136.01(5SF)	1	CRV	10174
17	225	0	Inf. Dilution	dG:136.18(5SF)	1	CRV	10174
18	250	0	Inf. Dilution	dG:136.31(5SF)	0	CRV	10174
19	300	0	Inf. Dilution	dG:136.41(5SF)	0	CRV	10174
20	23	0	Inf. Dilution	E:-2.92(3SF)	0	ENS	7276

Note (20): Suspect other values

21	25	0	Inf. Dilution	E:-2.92(0.01SD)	0	MDG	775
22	25	0	Inf. Dilution	E:-2.80(3SF)	0	CRV	7761

Note (22): Low wgt for internal consistency

23	25	0	Inf. Dilution	E:-2.916(4SF)	2	CRV	22551
24	25	0	Inf. Dilution	dH:126.1(0.1SD)	3	ENS	3006
25	25	0	Inf. Dilution	dH:126.1(4SF)	3	CRV	5223
26	25	0	Inf. Dilution	dH:126.1(4SF)	4	CRV	6488
27	25	0	Inf. Dilution	dH:515.0(4SF) kJ	0	CRV	7761
28	25	0	Inf. Dilution	dH:126.100(6SF)	4	CRV	9029
29	25	0	Inf. Dilution	dH:527.6(4SF) kJ	4	EFE	32151

Reaction No. 10650

Tartaric-2 + Ra+2 = Ra+2_Tartaric-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.16	NaCl	lgK:1.24(3SF)	5	MIX	6800
2	25	0.16	Unknown	lgK:1.24(3SF)	5	MIX	108

Reaction No. 20092

Citric-3 + Ra+2 = Ra+2_Citric-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	7	0.16	NaCl	lgK:2.0(2SF)	5	MIX	6800
2	25	0	Inf. Dilution	lgK:3.443(4SF)	1	EOR	

3	25	0	Inf. Dilution	lgK:3.59 (3SF)	4	CRV	31301
4	25	0.16	NaCl	lgK:2.05 (3SF)	5	MIX	6800
5	25	0.16	Unknown	lgK:2.36 (3SF)	3	MIX	108
6	37	0.16	NaCl	lgK:1.96 (3SF)	5	MIX	6800

Reaction No. 34746

Fumaric-2 + Ra+2 = Ra+2_Fumaric-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.16	NaCl	lgK:1.60 (3SF)	4	MIX	6800

Reaction No. 35551

Pyruvic-1 + Ra+2 = Ra+2_Pyruvic-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.16	NaCl	lgK:0.89 (2SF)	4	MIX	6800
2	25	0.16	Unknown	lgK:1.9 (2SF)	3	MIX	140

Reaction No. 35571

Succinic-2 + Ra+2 = Ra+2_Succinic-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.16	NaCl	lgK:1.0 (2SF)	4	MIX	6800

Reaction No. 44957

HOEDTA-3 + Ra+2 = Ra+2_HOEDTA-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:5.6 (2SF)	4	MDS	906

Reaction No. 45253

CDTA-4 + Ra+2 = Ra+2_CDTA-4

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:8.3 (2SF)	4	MDS	906

Reaction No. 45277

DTPA-5 + Ra+2 = Ra+2_DTPA-5

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:8.5 (2SF)	4	MDS	906
2	25	0.1	NaClO4	lgK:8.5 (0.1SD)	5	CRV	13242

Reaction No. 45293

EGTA-4 + Ra+2 = Ra+2_EGTA-4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:7.7(2SF)	4	MDS	906

Reaction No. 46028							
Asp-2 + Ra+2 = Ra+2_Asp-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KCl	lgK:0.86(2SF)	3	MGL	6710
2	25	0.16	NaCl	lgK:0.85(2SF)	5	MIX	6800

Reaction No. 55557							
RaO(s) + 2<H+1> + 2<e-1> = Ra(s) + H2O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:-1.32(0.01SD)	5	ENS	3006

Reaction No. 55558							
Ra(s) + N2(g) + 3<O2(g)> = Ra+2_NO3-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-187.4(0.1SD)	5	ENS	3006
2	25	0	Inf. Dilution	dH:-225.2(0.1SD)	5	ENS	3006

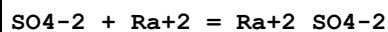
Reaction No. 55708							
Ra(s) + C(s) + 1.5<O2(g)> = Ra+2_CO3-2_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-271.69(5SF)	4	CRV	5223
2	25	0	Inf. Dilution	dH:-290.73(5SF)	3	CRV	5223

Reaction No. 55709							
Ra(s) + 0.5<Cl2(g)> = Ra+2_Cl-1 + e-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-165.44(5SF)	4	CRV	5223
2	25	0	Inf. Dilution	dH:-165.54(5SF)	4	CRV	5223

Reaction No. 55710							
Ra(s) + 0.5<O2(g)> + 0.5<H2(g)> = Ra+2_OH-1 + e-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-172.3(5SF)	4	CRV	5223

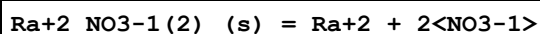
2	25	0	Inf. Dilution	dH:-179.98 (5SF)	4	CRV	5223
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Reaction No. 56071



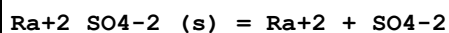
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.754 (4SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:2.75 (3SF)	4	CRV	5223
3	25	0	Inf. Dilution	lgK:1.24 (3SF)	0	EBU	6372
4	25	0	Inf. Dilution	lgK:2.75 (3SF)	4	ENS	6496
5	25	0	Inf. Dilution	lgK:2.76 (0.05SD)	5	EOD	22958
6	25	0	Inf. Dilution	lgK:2.72 (0.045SD)	4	EOD	31301
7	25	0	CHEMVAL4	lgK:0.02 (0.01SD)	0	ENS	5156
8	25	0	Inf. Dilution	dH:1.3 (2SF)	4	ENS	6496

Reaction No. 56307



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-2.05 (3SF)	2	CRV	32224
2	25	0	CHEMVAL4	lgK:-3.00 (0.01SD)	3	ENS	5156

Reaction No. 56308



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-10.31 (4SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:-10.37 (4SF)	4	CRV	552
3	25	0	Inf. Dilution	lgK:-10.38 (0.02SD)	4	CRV	5223
4	25	0	Inf. Dilution	lgK:-10.26 (4SF)	4	ENS	6496
5	25	0	Inf. Dilution	lgK:-10.21 (0.06SD)	5	CRV	8723
6	25	0	Inf. Dilution	lgK:-14.38 (4SF)	0	CRV	22551
7	25	0	Inf. Dilution	lgK:-10.05 (0.2SD)	3	EOD	31301
8	25	0	Inf. Dilution	lgK:-10.43 (4SF)	4	CRV	32224
9	25	0:4	Inf. Dilution	lgK:-10.4 (3SF)	0	EOD	22544
10	25	0	CHEMVAL4	lgK:-10.40 (0.01SD)	4	ENS	5156
11	35	0	Inf. Dilution	lgK:-10.05 (4SF)	3	EOD	8723
12	45	0	Inf. Dilution	lgK:-9.87 (3SF)	3	EOD	8723
13	60	0	Inf. Dilution	lgK:-9.66 (3SF)	3	EOD	8723
14	25	0	Inf. Dilution	dH:9.40 (3SF)	4	ENS	6496

Reaction No. 56309							
$\text{Ra+2_Cl-1(2)}_{\text{(s)}} = \text{Ra+2} + 2\text{<Cl-1>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	CHEMVAL4	$\lg K: -0.80 (0.01\text{SD})$	4	ENS	5156

Reaction No. 57193							
$\text{Ra(s)} + \text{N}_2(\text{g}) + 3\text{<O}_2(\text{g})\text{>} = \text{Ra+2_NO}_3\text{-1(2)}_{\text{(s)}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\text{dG}: -190.3 (0.1\text{SD})$	5	ENS	3006
2	25	0	Inf. Dilution	$\text{dH}: -237.1 (0.1\text{SD})$	5	ENS	3006

Reaction No. 57194							
$\text{Ra(s)} + \text{S(s)} + 2\text{<O}_2(\text{g})\text{>} = \text{Ra+2_SO}_4\text{-2}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\text{dG}: -312.2 (0.1\text{SD})$	0	ENS	3006
2	25	0	Inf. Dilution	$\text{dG}: -315.90 (5\text{SF})$	4	CRV	5223
3	25	0	Inf. Dilution	$\text{dH}: -343.4 (0.1\text{SD})$	3	ENS	3006
4	25	0	Inf. Dilution	$\text{dH}: -342.18 (5\text{SF})$	3	CRV	5223

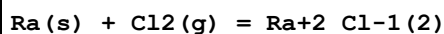
Reaction No. 57195							
$\text{Ra(s)} + \text{S(s)} + 2\text{<O}_2(\text{g})\text{>} = \text{Ra+2_SO}_4\text{-2}_{\text{(s)}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\text{dG}: -326.4 (0.1\text{SD})$	4	ENS	3006
2	25	0	Inf. Dilution	$\text{dG}: -326.15 (5\text{SF})$	4	CRV	5223
3	25	0	Inf. Dilution	$\text{dH}: -351.6 (0.1\text{SD})$	3	ENS	3006
4	25	0	Inf. Dilution	$\text{dH}: -352.88 (5\text{SF})$	3	CRV	5223

Reaction No. 57196							
$\text{Ra(s)} + \text{I}_2(\text{s}) + 3\text{<O}_2(\text{g})\text{>} = \text{Ra+2_IO}_3\text{-1(2)}_{\text{(s)}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\text{dG}: -207.6 (0.1\text{SD})$	5	ENS	3006
2	25	0	Inf. Dilution	$\text{dH}: -245.2 (0.1\text{SD})$	5	ENS	3006

Reaction No. 57197							
$\text{Ra(s)} + \text{Cl}_2(\text{g}) + 2\text{<H}_2(\text{g})\text{>} + \text{O}_2(\text{g}) = \text{Ra+2_Cl-1(2)_H}_2\text{O(2)}_{\text{(s)}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

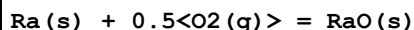
1	25	0	Inf. Dilution	dG:-311.4(0.1SD)	5	ENS	3006
2	25	0	Inf. Dilution	dH:-349.9(0.1SD)	5	ENS	3006

Reaction No. 57198



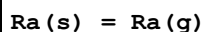
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1	25	0	Inf. Dilution	dG:-196.9(0.1SD)	5	ENS	3006
2	25	0	Inf. Dilution	dH:-206.0(0.1SD)	5	ENS	3006

Reaction No. 57199



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-117.5(0.1SD)	5	ENS	5551
2	25	0	Inf. Dilution	dH:-125.(0.5SD)	4	ENS	3006

Reaction No. 57202



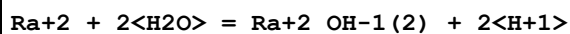
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:31.1(0.1SD)	5	ENS	3006
2	25	0	Inf. Dilution	dH:38.(0.5SD)	4	ENS	3006
3	25	0	Inf. Dilution	dH:155.(3SF) kJ	3	CRV	22524

Reaction No. 64649



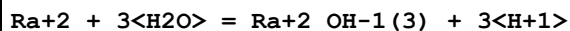
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1	25	0	Inf. Dilution	lgK:-13.35(4SF)	1	EBU	6372
2	25	0	Inf. Dilution	lgK:-13.49(4SF)	3	ENS	6496
3	25	0	Inf. Dilution	lgK:-13.47(0.25SD)	4	EOD	31301

Reaction No. 64650



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-28.01(4SF)	1	EBU	6372

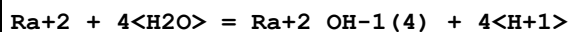
Reaction No. 64651



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
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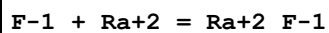
1	25	0	Inf. Dilution	lgK:-43.83(4SF)	1	EBU	6372
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Reaction No. 64652



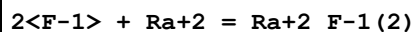
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1	25	0	Inf. Dilution	lgK:-60.78(4SF)	1	EBU	6372

Reaction No. 64653



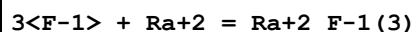
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Reaction No. 64654



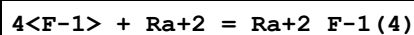
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.50(2SF)	1	EBU	6372

Reaction No. 64655



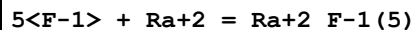
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1	25	0	Inf. Dilution	lgK:0.19(2SF)	1	EBU	6372

Reaction No. 64656



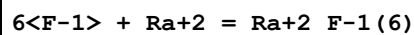
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Reaction No. 64657



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-1.28(3SF)	1	EBU	6372

Reaction No. 64658



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-2.38(3SF)	1	EBU	6372

Reaction No. 64659

Cl-1 + Ra+2 = Ra+2_Cl-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.1(1SF)	3	CRV	5223
2	25	0	Inf. Dilution	lgK:0.16(2SF)	0	EBU	6372

Reaction No. 64660							
2<Cl-1> + Ra+2 = Ra+2_Cl-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.35(2SF)	1	EBU	6372

Reaction No. 64661							
3<Cl-1> + Ra+2 = Ra+2_Cl-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-1.41(3SF)	1	EBU	6372

Reaction No. 64662							
4<Cl-1> + Ra+2 = Ra+2_Cl-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-2.96(3SF)	1	EBU	6372

Reaction No. 64663							
5<Cl-1> + Ra+2 = Ra+2_Cl-1(5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-4.99(3SF)	1	EBU	6372

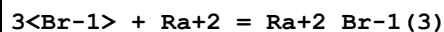
Reaction No. 64664							
6<Cl-1> + Ra+2 = Ra+2_Cl-1(6)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-7.46(3SF)	1	EBU	6372

Reaction No. 64665							
Br-1 + Ra+2 = Ra+2_Br-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.14(2SF)	1	EBU	6372

Reaction No. 64666							
2<Br-1> + Ra+2 = Ra+2_Br-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

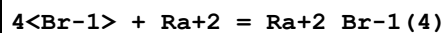
1	25	0	Inf. Dilution	lgK:-0.15 (2SF)	1	EBU	6372
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Reaction No. 64667



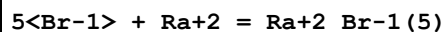
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.76 (2SF)	1	EBU	6372

Reaction No. 64668



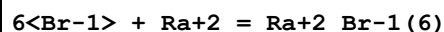
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-1.62 (3SF)	1	EBU	6372

Reaction No. 64669



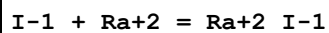
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-2.71 (3SF)	1	EBU	6372

Reaction No. 64670



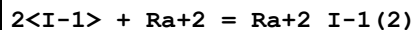
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-4.01 (3SF)	1	EBU	6372

Reaction No. 64671



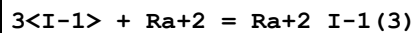
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.13 (2SF)	1	EBU	6372

Reaction No. 64672



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.07 (1SF)	1	EBU	6372

Reaction No. 64673



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.45 (2SF)	1	EBU	6372

Reaction No. 64674

4<I-1> + Ra+2 = Ra+2_I-1 (4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.98 (2SF)	1	EBU	6372

Reaction No. 64675							
5<I-1> + Ra+2 = Ra+2_I-1 (5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-1.62 (3SF)	1	EBU	6372

Reaction No. 64676							
6<I-1> + Ra+2 = Ra+2_I-1 (6)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-2.36 (3SF)	1	EBU	6372

Reaction No. 64677							
IO3-1 + Ra+2 = Ra+2_IO3-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.25 (2SF)	1	EBU	6372

Reaction No. 64678							
2<IO3-1> + Ra+2 = Ra+2_IO3-1 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.82 (2SF)	1	EBU	6372

Reaction No. 64679							
3<IO3-1> + Ra+2 = Ra+2_IO3-1 (3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-2.95 (3SF)	1	EBU	6372

Reaction No. 64680							
2<SO4-2> + Ra+2 = Ra+2_SO4-2 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.23 (3SF)	1	EBU	6372

Reaction No. 64681							
3<SO4-2> + Ra+2 = Ra+2_SO4-2 (3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.20 (2SF)	1	EBU	6372

Reaction No. 64682							
$4\text{SO}_4^{2-} + \text{Ra}^{2+} = \text{Ra}^{2+}\text{SO}_4^{2-} (4)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -1.72 (3\text{SF})$	1	EBU	6372

Reaction No. 64683							
$5\text{SO}_4^{2-} + \text{Ra}^{2+} = \text{Ra}^{2+}\text{SO}_4^{2-} (5)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -4.50 (3\text{SF})$	1	EBU	6372

Reaction No. 64684							
$\text{NH}_3 + \text{Ra}^{2+} = \text{Ra}^{2+}\text{NH}_3$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -0.76 (2\text{SF})$	1	EBU	6372

Reaction No. 64685							
$2\text{NH}_3 + \text{Ra}^{2+} = \text{Ra}^{2+}\text{NH}_3 (2)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -2.78 (3\text{SF})$	1	EBU	6372

Reaction No. 64686							
$\text{NO}_3^- + \text{Ra}^{2+} = \text{Ra}^{2+}\text{NO}_3^-$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: 0.19 (2\text{SF})$	1	EBU	6372

Reaction No. 64687							
$2\text{NO}_3^- + \text{Ra}^{2+} = \text{Ra}^{2+}\text{NO}_3^- (2)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: 0.2 (1\text{SF})$	1	ELC	
2	25	0	Inf. Dilution	$\lg K: -1.15 (3\text{SF})$	0	EBU	6372

Reaction No. 64688							
$3\text{NO}_3^- + \text{Ra}^{2+} = \text{Ra}^{2+}\text{NO}_3^- (3)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -3.75 (3\text{SF})$	1	EBU	6372

Reaction No. 64689							
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Ra+2 + H+1_PO4-3 = Ra+2_H+1_PO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.94(3SF)	1	EBU	6372

Reaction No. 64690							
Ra+2 + 2<H+1_PO4-3> = Ra+2_H+1(2)_PO4-3(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.01(3SF)	1	EBU	6372

Reaction No. 64691							
Ra+2 + 3<H+1_PO4-3> = Ra+2_H+1(3)_PO4-3(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.48(3SF)	1	EBU	6372

Reaction No. 64692							
Ra+2 + 4<H+1_PO4-3> = Ra+2_H+1(4)_PO4-3(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.44(3SF)	1	EBU	6372

Reaction No. 64693							
Ra+2 + 5<H+1_PO4-3> = Ra+2_H+1(5)_PO4-3(5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.95(3SF)	1	EBU	6372

Reaction No. 64694							
Ra+2 + 6<H+1_PO4-3> = Ra+2_H+1(6)_PO4-3(6)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.04(3SF)	1	EBU	6372

Reaction No. 64695							
Ra+2 + H+1(2)_PO4-3 = Ra+2_H+1(2)_PO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.38(2SF)	1	EBU	6372

Reaction No. 64696							
Ra+2 + 2<H+1(2)_PO4-3> = Ra+2_H+1(4)_PO4-3(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.16(2SF)	1	EBU	6372

Reaction No. 64697							
$\text{Ra}+2 + 3\text{<H+1(2)_PO4-3>} = \text{Ra}+2\text{_H+1(6)_PO4-3(3)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -1.37(3\text{SF})$	1	EBU	6372

Reaction No. 64698							
$\text{Ra}+2 + 4\text{<H+1(2)_PO4-3>} = \text{Ra}+2\text{_H+1(8)_PO4-3(4)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -3.14(3\text{SF})$	1	EBU	6372

Reaction No. 64699							
$\text{Ra}+2 + 5\text{<H+1(2)_PO4-3>} = \text{Ra}+2\text{_H+1(10)_PO4-3(5)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -5.44(3\text{SF})$	1	EBU	6372

Reaction No. 64700							
$\text{Ra}+2 + 6\text{<H+1(2)_PO4-3>} = \text{Ra}+2\text{_H+1(12)_PO4-3(6)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -8.21(3\text{SF})$	1	EBU	6372

Reaction No. 64701							
$\text{CO3-2} + \text{Ra}+2 = \text{Ra}+2\text{_CO3-2}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: 2.5(2\text{SF})$	4	CRV	5223
2	25	0	Inf. Dilution	$\lg K: 2.82(3\text{SF})$	0	EBU	6372
3	25	0	Inf. Dilution	$\lg K: 2.5(2\text{SF})$	4	ENS	6496
4	25	0	Inf. Dilution	$\text{dH}: 1.07(3\text{SF})$	4	ENS	6496

Reaction No. 64702							
$2\text{<CO3-2>} + \text{Ra}+2 = \text{Ra}+2\text{_CO3-2(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: 4.05(3\text{SF})$	1	EBU	6372

Reaction No. 64703							
$3\text{<CO3-2>} + \text{Ra}+2 = \text{Ra}+2\text{_CO3-2(3)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: 3.94(3\text{SF})$	1	EBU	6372

Reaction No. 64704							
$4\text{<CO3-2>} + \text{Ra+2} = \text{Ra+2_CO3-2 (4)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.58 (3SF)	1	EBU	6372

Reaction No. 64705							
$5\text{<CO3-2>} + \text{Ra+2} = \text{Ra+2_CO3-2 (5)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.04 (1SF)	1	EBU	6372

Reaction No. 64706							
$\text{Ra+2} + \text{H+1_CO3-2} = \text{Ra+2_H+1_CO3-2}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.42 (2SF)	1	EBU	6372

Reaction No. 64707							
$\text{Ra+2} + 2\text{<H+1_CO3-2>} = \text{Ra+2_H+1 (2)_CO3-2 (2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.67 (2SF)	1	EBU	6372

Reaction No. 64708							
$\text{Ra+2} + 3\text{<H+1_CO3-2>} = \text{Ra+2_H+1 (3)_CO3-2 (3)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-3.01 (3SF)	1	EBU	6372

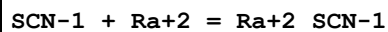
Reaction No. 64709							
$\text{CN-1} + \text{Ra+2} = \text{Ra+2_CN-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.50 (2SF)	1	EBU	6372

Reaction No. 64710							
$2\text{<CN-1>} + \text{Ra+2} = \text{Ra+2_CN-1 (2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.27 (2SF)	1	EBU	6372

Reaction No. 64711							
$3\text{<CN-1>} + \text{Ra+2} = \text{Ra+2_CN-1 (3)}$							

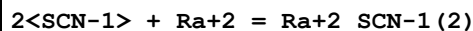
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-2.21 (3SF)	1	EBU	6372

Reaction No. 64712



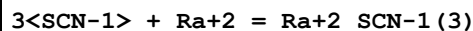
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.21 (2SF)	1	EBU	6372

Reaction No. 64713



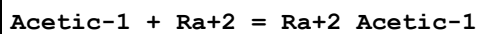
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.77 (2SF)	1	EBU	6372

Reaction No. 64714



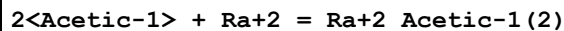
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-2.84 (3SF)	1	EBU	6372

Reaction No. 64715



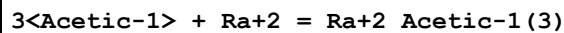
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.37 (2SF)	1	EBU	6372

Reaction No. 64716



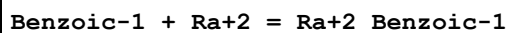
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.81 (2SF)	1	EBU	6372

Reaction No. 64717



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-3.30 (3SF)	1	EBU	6372

Reaction No. 64718



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.40 (2SF)	1	EBU	6372

Reaction No. 64719							
2<Benzoic-1> + Ra+2 = Ra+2_Benzoic-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.80(2SF)	1	EBU	6372

Reaction No. 64720							
3<Benzoic-1> + Ra+2 = Ra+2_Benzoic-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-3.35(3SF)	1	EBU	6372

Reaction No. 64721							
Cat-2 + Ra+2 = Ra+2_Cat-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.33(3SF)	1	EBU	6372

Reaction No. 64722							
2<Cat-2> + Ra+2 = Ra+2_Cat-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:6.64(3SF)	1	EBU	6372

Reaction No. 64723							
3<Cat-2> + Ra+2 = Ra+2_Cat-2(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.61(3SF)	1	EBU	6372

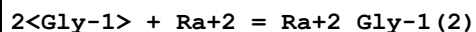
Reaction No. 64724							
EtDiAm + Ra+2 = Ra+2_EtDiAm							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.05(1SF)	1	EBU	6372

Reaction No. 64725							
2<EtDiAm> + Ra+2 = Ra+2_EtDiAm(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-1.98(3SF)	1	EBU	6372

Reaction No. 64726							
Gly-1 + Ra+2 = Ra+2_Gly-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

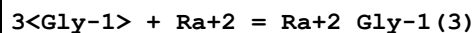
1	25	0	Inf. Dilution	lgK:0.31(2SF)	1	EBU	6372
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Reaction No. 64727



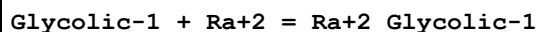
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.95(2SF)	1	EBU	6372

Reaction No. 64728



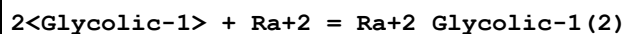
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-2.82(3SF)	1	EBU	6372

Reaction No. 64729



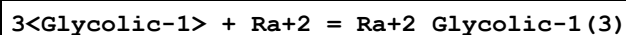
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.34(2SF)	1	EBU	6372

Reaction No. 64730



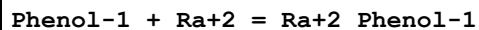
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.83(2SF)	1	EBU	6372

Reaction No. 64731



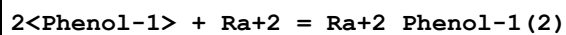
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-3.25(3SF)	1	EBU	6372

Reaction No. 64732



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.52(2SF)	1	EBU	6372

Reaction No. 64733



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.23(2SF)	1	EBU	6372

Reaction No. 64734

3<Phenol-1> + Ra+2 = Ra+2_Phenol-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-2.14(3SF)	1	EBU	6372

Reaction No. 65518							
Ra+2 + 2<H2O> = Ra+2_OH-1(2)_(s) + 2<H+1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-21.06(4SF)	1	EBU	6372

Reaction No. 65519							
Ra+2 + CO3-2 = Ra+2_CO3-2_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.3(2SF)	3	CRV	5223
2	25	0	Inf. Dilution	lgK:9.15(3SF)	0	EBU	6372
3	25	0	Inf. Dilution	lgK:8.54(0.10SD)	4	EOD	31301

Reaction No. 68607							
OxAcet-2 + Ra+2 = Ra+2_OxAcet-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.16	NaCl	lgK:1.82(3SF)	5	MIX	6800

Reaction No. 68608							
Ra+2 + H+1_SulfSal-3 = Ra+2_H+1_SulfSal-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.16	NaCl	lgK:1.92(3SF)	5	MIX	6800

Reaction No. 69732							
Malic-2 + Ra+2 = Ra+2_Malic-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.16	Unknown	lgK:0.95(2SF)	4	MIX	108

Reaction No. 71629							
Ra+2 + H2(g) + 2<e-1> = RaH2(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:-2.19(3SF)	3	CRV	7761
2	25	0	Inf. Dilution	dH:359.9(4SF) kJ	3	CRV	7761

Reaction No. 71630							
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Ra+2_OH-1 + H+1 + 2<e-1> = Ra(s) + H2O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:-2.40 (3SF)	0	CRV	7761

Reaction No. 71631							
Ra+2_OH-1 + 2<e-1> = Ra(s) + OH-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:-2.81 (3SF)	0	CRV	7761
Note (1): Low wgt for inter-reaction consistency							

Reaction No. 71632							
Ra+2_OH-1(2)_H2O(8)_ (s) + 2<e-1> = Ra(s) + 2<OH-1> + 8<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:-2.89 (3SF)	3	CRV	7761

Reaction No. 73883							
C(s) + 1.5<O2(g)> + Ra(s) = Ra+2_CO3-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-263.78 (5SF)	4	CRV	5223
2	25	0	Inf. Dilution	dH:-286.87 (5SF)	3	CRV	5223

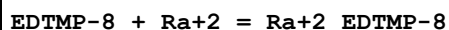
Reaction No. 73884							
Ra+2 + OH-1 = Ra+2_OH-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.5 (1SF)	3	CRV	5223

Reaction No. 73978							
Oxalic-2 + Ra+2 = Ra+2_Oxalic-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.2 (2SF)	2	EES	
Note (1): Estimate based on Ba+2 comparability							
2	25	0	Inf. Dilution	lgK:2.2 (4SF)	1	EOR	
3	25	0	Inf. Dilution	lgK:2.78 (3SF)	4	CRV	31301

Reaction No. 73979							
NTA-3 + Ra+2 = Ra+2_NTA-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

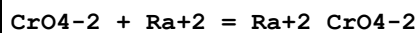
1	25	0	Inf. Dilution	lgK:5.7 (2SF)	2	EES	
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Reaction No. 73980



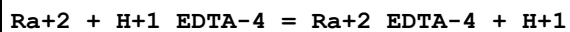
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:10.6 (3SF)	2	EES	

Reaction No. 75005



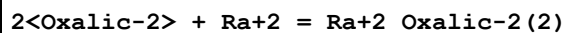
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	0.01	0	Inf. Dilution	lgK:3.1567 (5SF)	4	EOD	22958
2	25	0	Inf. Dilution	lgK:3.0215 (5SF)	4	EOD	22958
3	60	0	Inf. Dilution	lgK:2.9559 (5SF)	4	EOD	22958
4	100	0	Inf. Dilution	lgK:2.9836 (5SF)	4	EOD	22958
5	150	0	Inf. Dilution	lgK:3.1423 (5SF)	4	EOD	22958
6	200	0	Inf. Dilution	lgK:3.4477 (5SF)	4	EOD	22958

Reaction No. 80033



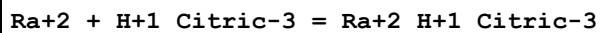
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	NaCl/m	lgK:-2.07 (0.11SD)	3	MIX	29889
2	25	0.22	NaCl/m	lgK:-3.26 (0.06SD)	3	MIX	29889
3	25	0.51	NaCl/m	lgK:-3.42 (0.08SD)	3	MIX	29889
4	25	1.02	NaCl/m	lgK:-3.85 (0.08SD)	3	MIX	29889
5	25	2.09	NaCl/m	lgK:-4.15 (0.10SD)	3	MIX	29889
6	25	2.64	NaCl/m	lgK:-4.24 (0.08SD)	3	MIX	29889

Reaction No. 80625



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.44 (3SF)	4	CRV	31301

Reaction No. 80629



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.64 (3SF)	4	CRV	31301

Reaction No. 81564							
$\text{Ra}+2\text{Cl-1}(2)\text{H}_2\text{O}(2)\text{(s)} = \text{Ra}+2 + 2\text{Cl-1} + 2\text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -0.80(2\text{SF})$	4	CRV	32224

Data Assessment

Weight	Assessment
9	highest accuracy
8	multi-determined; excellent dependability
7	trustworthy; outstanding single determination
6	better than average reliability
5	average single determination
4	some doubt
3	poorly known
2	guess to achieve consistency (or just as bad)
1	no information but consistent
0	problematic/inconsistent

Techniques

Abbrev	Method
CRV	Critical review
EBU	Brown's Unified Theory
EDH	Debye-Hueckel corrections
EES	Personal estimate
EFE	Free Energy relations
EIU	Critical IUPAC review (estim.)
ELC	Linear Combination of Reactions
ENS	Not specified
EOD	Calculated from data of others
EOR	Other Reaction Constants
MDG	Combination of Thermodynamic Data

MDS	Distribution between two phases
MGL	Glass electrode
MIX	Ion exchange
MSC	Miscellaneous

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